Computer Vision

HW 5: Simple Background Subtraction

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Introduction

Moving object segmentation is often an important application of computer vision whenever it comes to working with video. Since video allows for more context of the scene to be analyzed, a simple method of segmenting moving objects is subtracting the background from the regions that contain the foreground, though this only works with a non-moving camera. This paper will experiment with different learning rates and threshold values and their pros and cons.

Experiments and Results



Figure 1. Input frames 8, 120, 180, 260.

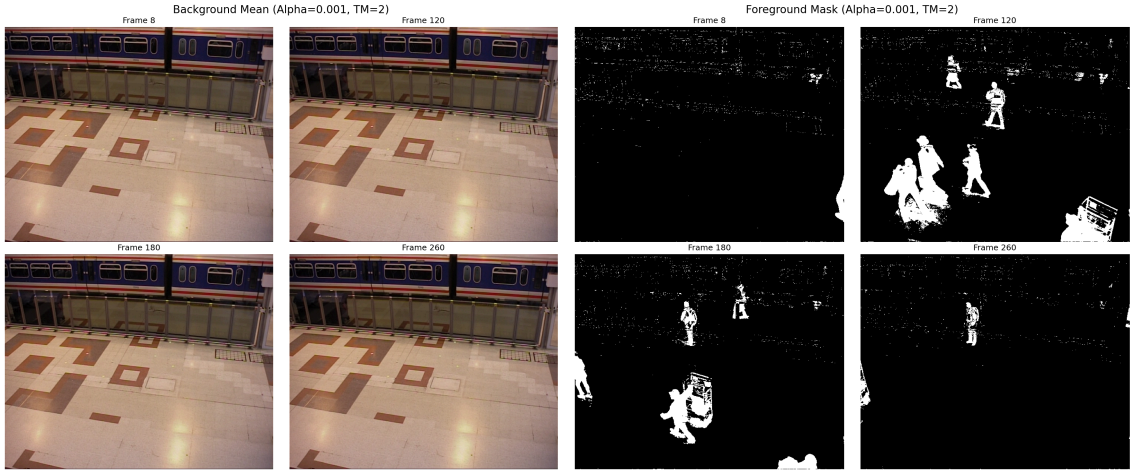


Figure 1. Background mean images and foreground mask images with Alpha = .001 and TM = 2.

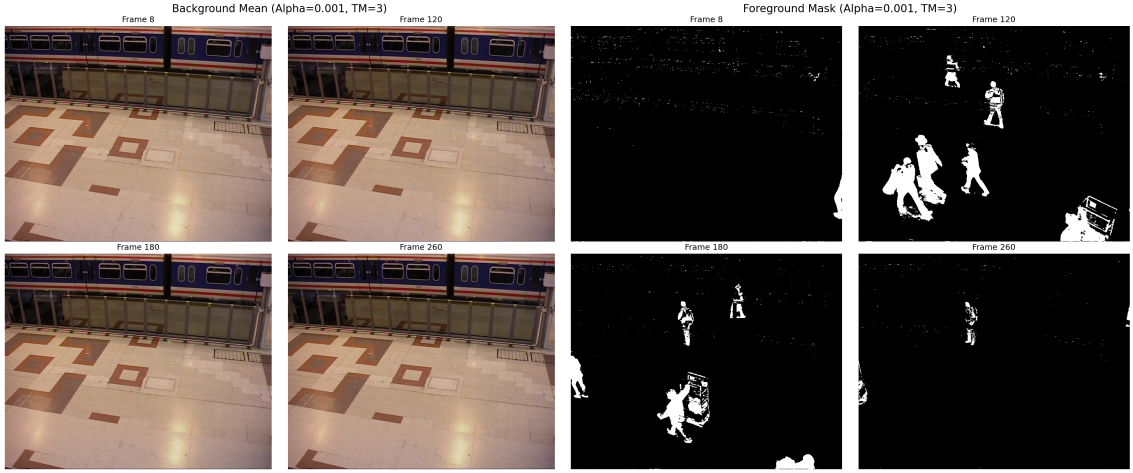


Figure 2. Background mean images and foreground mask images with Alpha = .001 and TM = 3.

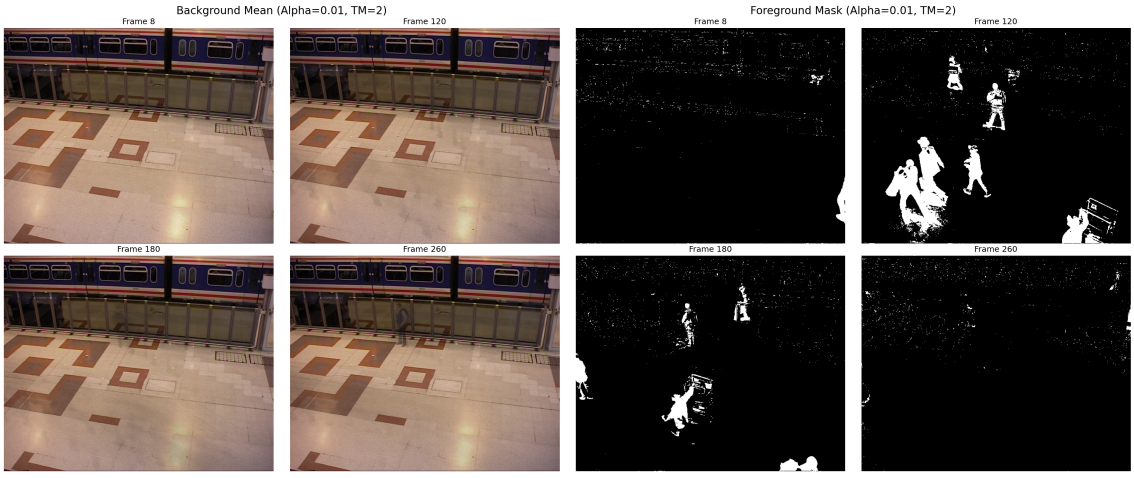


Figure 3. Background mean images and foreground mask images with Alpha = .01 and TM = 2.

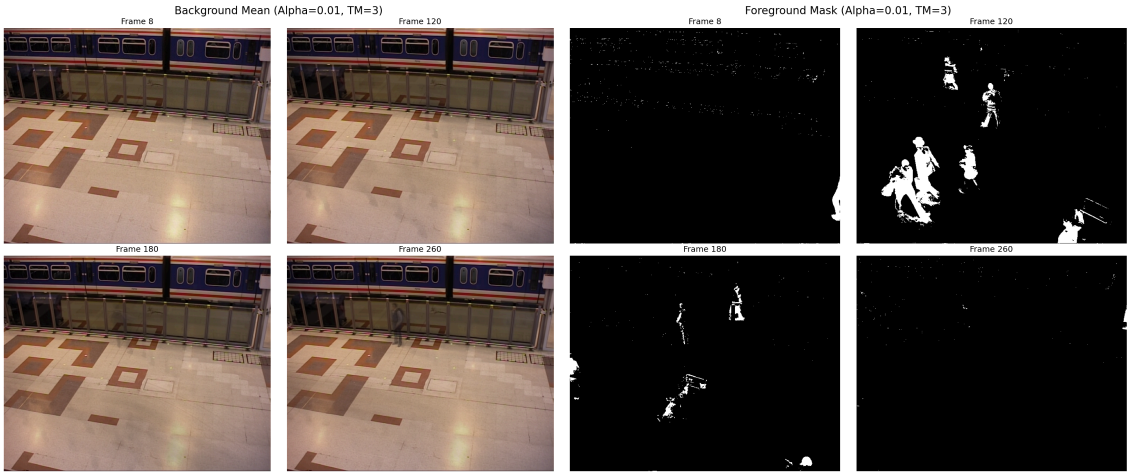


Figure 4. Background mean images and foreground mask images with Alpha = .01 and TM = 3.

Discussion

From the results, the first glaring issue seen within the foreground masks is the noise. Especially with the test where the TM = 2, there is a lot of noise towards the top of the images. Though with the images that are TM = 3, parts of the actual humans are being thresholded, making the object masking less accurate as well. A possible method to help fix this would be using post processing methods like dilation and erosion.

Seen in figures 3 and 4, where the learning rate is 0.01, the background mean is much less clean, where you can see some motion blur where the people were walking. This would be because the background model updates quicker, being more sensitive to recent moving objects in the frame. Overall it seems that the lower learning rate and higher threshold performed the best for balancing having less noise while keeping the moving objects intact for the foreground mask.